

Topic 07: Homogeneous First-Order Equations & Geometric Substitutions

Professor Leonard Master Series · Lessons 20 & 21

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Radial Transformation

Professor Leonard introduces homogeneous substitutions by looking at slope along rays:

"What if an equation isn't separable, but every single term has the exact same total algebraic degree? That means along any radial line from the origin ($y = vx$), the slope dy/dx is completely constant! If you change variables to the slope of that ray, $v = y/x$, the entire equation instantly collapses into a clean, separable ODE!"

A function $f(x, y)$ is **homogeneous of degree n** if:

$$f(tx, ty) = t^n f(x, y)$$

If an ODE can be written as $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$, substitute:

$$\begin{aligned} y = vx &\implies \frac{dy}{dx} = v + x \frac{dv}{dx} \\ v + x \frac{dv}{dx} &= F(v) \implies \frac{dv}{F(v) - v} = \frac{dx}{x} \end{aligned}$$

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 7.1: Homogeneous of Degree 2 (The Classic Circle Family)

Statement: Solve $(x^2 + y^2)dx - 2xydy = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Check Homogeneity:**
 - $M(tx, ty) = t^2x^2 + t^2y^2 = t^2(x^2 + y^2)$ (degree 2).
 - $N(tx, ty) = -2(tx)(ty) = t^2(-2xy)$ (degree 2). Both terms have degree 2. The equation is homogeneous!

- **Step 2: Express dy/dx as a function of y/x :**

$$2xydy = (x^2 + y^2)dx \implies \frac{dy}{dx} = \frac{x^2 + y^2}{2xy} = \frac{1 + (y/x)^2}{2(y/x)}$$

- **Step 3: Substitute $y = vx$ and $dy/dx = v + xdv/dx$:**

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

- **Step 4: Isolate xdv/dx :**

$$x \frac{dv}{dx} = \frac{1 + v^2}{2v} - v = \frac{1 + v^2 - 2v^2}{2v} = \frac{1 - v^2}{2v}$$

- **Step 5: Separate variables:**

$$\frac{2v}{1 - v^2} dv = \frac{dx}{x}$$

- **Step 6: Integrate both sides:** LHS: Let $u = 1 - v^2$, $du = -2v dv \implies -\ln|1 - v^2| = \ln|x| + C_1$.

$$\ln|1 - v^2|^{-1} = \ln|x| + C_1 \implies \frac{1}{|1 - v^2|} = C|x|$$

$$1 - v^2 = \frac{C_2}{x}$$

- **Step 7: Back-substitute $v = y/x$:**

$$1 - \frac{y^2}{x^2} = \frac{C_2}{x} \implies \frac{x^2 - y^2}{x^2} = \frac{C_2}{x}$$

Multiply by x^2 :

$$x^2 - y^2 = C_2 x \implies x^2 - C_2 x - y^2 = 0$$

This represents a family of hyperbolas passing through the origin!

Problem 7.2: Radical Homogeneous Equation & Arc-Sine

Statement: Solve $x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}$ for $x > 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Divide by x to create y/x terms:**

$$\frac{dy}{dx} = \frac{y}{x} + \frac{\sqrt{x^2 - y^2}}{x} = \frac{y}{x} + \sqrt{1 - \left(\frac{y}{x}\right)^2}$$

- **Step 2: Substitute** $y = vx$, $dy/dx = v + xdv/dx$:

$$v + x \frac{dv}{dx} = v + \sqrt{1 - v^2}$$

- **Step 3: Subtract v from both sides:**

$$x \frac{dv}{dx} = \sqrt{1 - v^2}$$

- **Step 4: Separate:**

$$\frac{dv}{\sqrt{1 - v^2}} = \frac{dx}{x}$$

- **Step 5: Integrate:**

$$\arcsin(v) = \ln(x) + C \implies v(x) = \sin(\ln x + C)$$

- **Step 6: Back-substitute** $v = y/x$:

$$\frac{y}{x} = \sin(\ln x + C) \implies y(x) = x \sin(\ln x + C)$$

Problem 7.3: Linear Homogeneous Ratio

Statement: Solve $\frac{dy}{dx} = \frac{x+3y}{3x+y}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Divide top and bottom by x :**

$$\frac{dy}{dx} = \frac{1 + 3(y/x)}{3 + (y/x)}$$

- **Step 2: Substitute** $y = vx$:

$$v + x \frac{dv}{dx} = \frac{1 + 3v}{3 + v}$$

$$x \frac{dv}{dx} = \frac{1 + 3v}{3 + v} - v = \frac{1 + 3v - 3v - v^2}{3 + v} = \frac{1 - v^2}{3 + v}$$

- **Step 3: Separate:**

$$\frac{3 + v}{1 - v^2} dv = \frac{dx}{x}$$

- **Step 4: Partial Fractions on LHS:**

$$\frac{3 + v}{(1 - v)(1 + v)} = \frac{A}{1 - v} + \frac{B}{1 + v}$$

$$3 + v = A(1 + v) + B(1 - v).$$

- $v = 1 \implies 4 = 2A \implies A = 2.$
- $v = -1 \implies 2 = 2B \implies B = 1.$

$$\int \left(\frac{2}{1-v} + \frac{1}{1+v} \right) dv = -2 \ln |1-v| + \ln |1+v| = \ln \frac{|1+v|}{(1-v)^2}$$

- **Step 5: Equate to $\ln |x| + C$:**

$$\frac{1+v}{(1-v)^2} = Cx$$

- **Step 6: Replace $v = y/x$:**

$$\frac{1+y/x}{(1-y/x)^2} = Cx \implies \frac{\frac{x+y}{x}}{\frac{(x-y)^2}{x^2}} = Cx \implies \frac{x(x+y)}{(x-y)^2} = Cx$$

Divide by x :

$$x+y = C(x-y)^2$$

Problem 7.4: Translation of Coordinates to Remove Non-Zero Intercepts

Statement: Transform $\frac{dy}{dx} = \frac{x-y+1}{x+y-3}$ into a homogeneous equation.

Step-by-Step Whiteboard Solution:

- **Step 1: Find the intersection of the two lines:**
 - $x - y + 1 = 0 \implies y = x + 1$
 - $x + y - 3 = 0 \implies y = -x + 3$ $x + 1 = -x + 3 \implies 2x = 2 \implies x_0 = 1, y_0 = 2.$
- **Step 2: Define translated variables:** Let $X = x - 1 \implies dX = dx$. Let $Y = y - 2 \implies dY = dy$.
Then $x = X + 1$ and $y = Y + 2$.
- **Step 3: Substitute into the ODE:**

$$\frac{dY}{dX} = \frac{(X+1) - (Y+2) + 1}{(X+1) + (Y+2) - 3} = \frac{X-Y}{X+Y}$$

The constant offsets vanish completely! The equation is now homogeneous of degree 1 and solved with $Y = vX$!

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting the Product Rule for dy/dx :** Writing $dy/dx = dv/dx$ instead of $v + x \frac{dv}{dx}$.
- **Trap 2: Forgetting to Back-Substitute:** Leaving your final exam answer in terms of v . You must replace $v = y/x$!